Yide Zhang

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PROFESSIONAL EXPERIENCE

University of Colorado Boulder (CU Boulder), Boulder, Colorado USA
Assistant Professor of Electrical, Computer & Energy Engineering *2025–Present*
Assistant Professor of Biomedical Engineering *2025–Present*
Director of the Boulder Optical Laboratory for Translational Science (BOLTS) *2025–Present*

EDUCATION AND TRAINING

California Institute of Technology (Caltech), Pasadena, California USA
NIH K99/R00 Postdoctoral Fellow in Medical Engineering and Electrical Engineering *2024–2025*
NIH T32 Postdoctoral Fellow in Medical Engineering and Electrical Engineering *2023–2024*
Postdoctoral Scholar in Medical Engineering and Electrical Engineering *2019–2023*
Advisor: Lihong V. Wang, Ph.D.

University of Notre Dame, Notre Dame, Indiana USA
Doctor of Philosophy in Electrical Engineering *2014–2019*
Master of Science in Electrical Engineering *2014–2017*
Advisor: Scott S. Howard, Ph.D.
Dissertation: *Super-Sensitivity and Super-Resolution Quantitative Multiphoton Microscopy*

Huazhong University of Science and Technology, Wuhan, China
Bachelor of Engineering in Automation *with Highest Honors* *2010–2014*
Advisor: Zhigang Zeng, Ph.D.
Thesis: *Research on Memristor-Based Memory Networks and Computing Technology*

RESEARCH INTERESTS

Photoacoustic imaging, quantum imaging, ultrafast imaging, fluorescence imaging, machine learning.

JOURNAL PUBLICATIONS

J39 Xin Tong†, Zhe He†, **Yide Zhang**†, Wenyu Liu†, Chien-Ying Huang, and Lihong V. Wang, "Above-twofold quantum super-resolution microscopy enabled by multiple idler passes with entangled biphotons", *Science Advances*, vol. 12, no. 30, pp. eaea9457, July 2026, DOI: 10.1126/sciadv.aea9457. Featured News.

J38 Binglin Shen, Zhi Zeng, Haiyang Li, Zongyi Yin, Jiazi Yu, Zhenglin Li, Chao Zhang, **Yide Zhang**, Rui Hu, Junle Qu, and Liwei Liu, "Simultaneous dual-Raman-shift scanning-free stimulated Raman scattering microscopy for label-free biomolecular imaging ", *Nature Communications*, May 2026, DOI: 10.1038/s41467-026-73391-8.

J37 **Yide Zhang**, "Universal restoration of medical images", *Nature Computational Science*, vol. 6, no. 4, pp. 321-322, Apr. 2026, DOI: 10.1038/s43588-026-00975-1.

J36 Jigmi Basumatary, Yousuf Aborahama, Yang Zhang, **Yide Zhang**, Yushun Zeng, Cindy Z. Liu, Qifa Zhou, Lihong V. Wang, "High-speed volumetric dual-mode ultrasound and photoacoustic tomography with a single-element detector", *IEEE Transactions on Medical Imaging*, vol. 45, no. 5, pp. 2317-2328, May 2026, DOI: 10.1109/TMI.2026.3653667.

J35 Xiaofei Luo†, Rui Cao†, Peng Hu, Yilin Luo, Yushun Zeng, **Yide Zhang**, Manxiu Cui, Qifa Zhou, Geng Ku, Lihong V. Wang, "Multifocal optical-resolution photoacoustic microscopy with a masked single-element transducer", *IEEE Transactions on Medical Imaging*, vol. 45, no. 5, pp. 2015-2022, May 2026, DOI: 10.1109/TMI.2025.3643-618.

J34 Byullee Park†, Rui Cao†, Yilin Luo†, Cindy Liu†, Yushun Zeng, **Yide Zhang**, Qifa Zhou, Samuel Davis, Massimo D'Apuzzo, and Lihong V. Wang, "Rapid cancer diagnosis using deep learningpowered label-free subcellular-resolution photoacoustic histology", *Science Advances*, vol. 11, no. 47, pp. eadz1820, Nov. 2025, DOI: 10.1126/sciadv.adz1820.

J33 **Yide Zhang** and Lihong V. Wang, "Emerging photoacoustic imaging techniques for peripheral arterial disease", *Current Treatment Options in Cardiovascular Medicine*, vol. 27, no. 56, pp. 1-10, Aug. 2025, DOI: 10.1007/s11936-025-01113-2.

J32 Xin Tong†, Cindy Z. Liu†, Yilin Luo†, Li Lin†, Jessica Dzubnar, Marta Invernizzi, Stephanie Delos Santos, **Yide Zhang**, Rui Cao, Peng Hu, Junfu Zheng, Jaclene Torres, Armine Kasabyan, Lily L. Lai, Lisa D. Yee, and Lihong V. Wang, "Panoramic photoacoustic computed tomography with learning-based classification enhances breast lesion characterization," *Nature Biomedical Engineering*, vol. 10, no. 1, pp. 161-177, June 2025, DOI: 10.1038/s41551-025-01435-3. Featured News.

J31 Rui Cao, Yilin Luo, Jingjing Zhao, Yushun Zeng, **Yide Zhang**, Qifa Zhou, Adam de la Zerda, and Lihong V. Wang, "Optical-resolution parallel ultraviolet photoacoustic microscopy for slide-free histology", *Science Advances*, vol. 10, no. 50, pp. eado0518, Dec. 2024, DOI: 10.1126/sciadv.ado0518. Featured News.

J30 Xin Tong, **Yide Zhang**, and Lihong V. Wang, "Enhancing optical microscopy with quantum entanglement", *Optics and Photonics News*, vol. 35, no. 11, pp. 32-39, Nov. 2024, DOI: 10.1364/OPN.35.11.000032.

J29 Kang Yong Loh†, Lei S. Li†, Jingyue Fan†, Yi Yiing Goh, Weng Heng Liew, Samuel Davis, **Yide Zhang**, Kai Li, Jie Liu, Liangliang Liang, Minjun Feng, Ming Yang, Hang Zhang, Ping'an Ma, Guangxue Feng, Zhao Mu, Weibo Gao, Tze Chien Sum, Bin Liu, Jun Lin, Kui Yao, Lihong V. Wang, and Xiaogang Liu, "Sharp-peaked lanthanide nanocrystals for near-infrared photoacoustic multiplexed differential imaging", *Communications Materials*, vol. 5, no. 164, pp. 1-10, Aug. 2024, DOI: 10.1038/s43246-024-00605-1.

J28 **Yide Zhang**†, Zhe He†, Xin Tong†, David C. Garrett, Rui Cao, and Lihong V. Wang, "Quantum imaging of biological organisms through spatial and polarization entanglement", *Science Advances*, vol. 10, no. 10, pp. eadk1495, Mar. 2024, DOI: 10.1126/sciadv.adk1495. Featured News.

J27 Ruofan Cao, **Yide Zhang**, and Jessica Houston, "Editorial: Phasor analysis for fluorescence lifetime data", *Frontiers in Bioinformatics* (Editorial), vol. 4, no. 1375480, pp. 1-2, Feb. 2024, DOI: 10.3389/fbinf.2024.1375480.

J26 **Yide Zhang**†, Peng Hu†, Lei Li, Rui Cao, Anjul Khadria, Konstantin Maslov, Xin Tong, Yushun Zeng, Laiming Jiang, Qifa Zhou, and Lihong V. Wang, "Ultrafast longitudinal imaging of haemodynamics via single-shot volumetric photoacoustic tomography with a single-element detector", *Nature Biomedical Engineering*, vol. 8, no. 6, pp. 712-725, Nov. 2023, DOI: 10.1038/s41551-023-01149-4. Highlighted in *Nature Biomedical Engineering*'s News & Views. Curated in *Nature Outlook*: Medical Diagnostics. Featured News. Featured in NIH NIBIB's *Science Highlights*.

J25 **Yide Zhang**, "Deep learning-enhanced microscopy with extended depth-of-field", *Light: Science & Applications*, vol. 12, no. 284, pp. 1-3, Nov. 2023, DOI: 10.1038/s41377-023-01323-y.

J24 Zhe He†, **Yide Zhang**†, Xin Tong†, Lei Li, and Lihong V. Wang, "Quantum microscopy of cells at the Heisenberg limit", *Nature Communications*, vol. 14, no. 2441, pp. 1-8, Apr. 2023, DOI: 10.1038/s41467-023-38191-4. Highlighted in Caltech's *2023 Year in Review*. Featured News.

J23 Xin Tong†, Zhe He†, **Yide Zhang**†, Samuel Solomon, Li Lin, Qiyuan Song, and Lihong V. Wang, "Experimental full-domain mapping of quantum correlation in Clauser-Horne-Shimony-Holt scenarios", *Physical Review Applied*, vol. 19, no. 034049, pp. 1-16, Feb. 2023, DOI: 10.1103/PhysRevApplied.19.034049.

J22 Yogeshwar Nath Mishra†, Peng Wang†, Florian J. Bauer, **Yide Zhang**, Dag Hanstorp, Stefan Will, and Lihong V. Wang, "Single-pulse real-time billion-frames-per-second planar imaging of ultrafast nanoparticle-laser dynamics and temperature in flames", *Light: Science & Applications*, vol. 12, no. 47, pp. 1-12, Feb. 2023, DOI: 10.1038/s41377-023-01095-5. *Light: Science & Applications* Top Downloads. Featured News.

J21 Zhongtao Cheng, Chengmingyue Li, Anjul Khadria, **Yide Zhang**, and Lihong V. Wang, "High-gain and high-speed wavefront shaping through scattering media", *Nature Photonics*, vol. 17, no. 2, pp. 299-305, Jan. 2023, DOI: 10.1038/s41566-022-01142-4. Featured News.

J20 Li Lin†, Xin Tong†, Susana Cavallero, **Yide Zhang**, Shuai Na, Rui Cao, Tzung K. Hsiai, and Lihong V. Wang, "Non-invasive photoacoustic computed tomography of rat heart anatomy and function", *Light: Science & Applications*, vol. 12, no. 12, pp. 1-9, Jan. 2023, DOI: 10.1038/s41377-022-01053-7.

J19 Rui Cao†, Jingjing Zhao†, Lei Li, Lin Du, **Yide Zhang**, Yilin Luo, Laiming Jiang, Samuel Davis, Qifa Zhou, Adam de la Zerda, and Lihong V. Wang, "Optical-resolution photoacoustic microscopy with a needle-shaped beam", *Nature Photonics*, vol. 17, no. 1, pp. 89-95, Dec. 2022, DOI: 10.1038/s41566-022-01112-w. Highlighted by *Nature*. Featured News.

J18 Rui Cao, Scott D. Nelson, Samuel Davis, Yu Liang, Yilin Luo, **Yide Zhang**, Brooke Crawford, and Lihong V. Wang, "Label-free intraoperative histology of bone tissue via deep-learning-assisted ultraviolet photoacoustic microscopy", *Nature Biomedical Engineering*, vol. 7, no. 2, pp. 124-134, Sept. 2022, DOI: 10.1038/s41551-022-00940-z. Featured News.

J17 **Yide Zhang**†, Binglin Shen†, Tong Wu, Jerry Zhao, Joseph C. Jing, Peng Wang, Kanomi Sasaki-Capela, William G. Dunphy, David Garrett, Konstantin Maslov, Weiwei Wang, and Lihong V. Wang, "Ultrafast and hypersensitive phase imaging of propagating internodal current flows in myelinated axons and electromagnetic pulses in dielectrics", *Nature Communications*, vol. 13, no. 5247, pp. 1-12, Sept. 2022, DOI: 10.1038/s41467-022-33002-8. Featured News.

J16 Varun Mannam, **Yide Zhang**, Yinhao Zhu, Evan Nichols, Qingfei Wang, Vignesh Sundaresan, Siyuan Zhang, Cody Smith, Paul W. Bohn, and Scott S. Howard, "Real-time image denoising of mixed Poisson-Gaussian noise in fluorescence microscopy images using ImageJ", *Optica*, vol. 9, no. 4, pp. 335-345, Apr. 2022, DOI: 10.1364/OPTICA.448287.

J15 **Yide Zhang**, Ian H. Guldner, Evan L. Nichols, David Benirschke, Cody J. Smith, Siyuan Zhang, and Scott S. Howard, "Instant FLIM enables 4D *in vivo* lifetime imaging of intact and injured zebrafish and mouse brains", *Optica*, vol. 8, no. 6, pp. 885-897, June 2021, DOI: 10.1364/OPTICA.426870. *Optica* Top Downloads.

J14 Lei Li†, Yang Li†, **Yide Zhang**, and Lihong V. Wang, "Snapshot photoacoustic topography through an ergodic relay of optical absorption *in vivo*", *Nature Protocols*, vol. 16, no. 5, pp. 2381-2394, May 2021, DOI: 10.1038/s41596-020-00487-w. *Nature Protocols*' Featured Protocol of the Week.

J13 Varun Mannam, **Yide Zhang**, Xiaotong Yuan, Cara Ravasio, and Scott S. Howard, "Machine learning for faster and smarter fluorescence lifetime imaging microscopy", *Journal of Physics: Photonics*, vol. 2, no. 4, pp. 042005, Sept. 2020, DOI: 10.1088/2515-7647/abac1a.

J12 **Yide Zhang**, Sergei Rouvimov, Xiaotong Yuan, Karla Gonzalez-Serrano, Alan C. Seabaugh, and Scott S. Howard, "Resolution enhancement of transmission electron microscopy by super-resolution radial fluctuations", *Applied Physics Letters*, vol. 116, no. 4, pp. 044105, Jan. 2020, DOI: 10.1063/1.5128353.

J11 **Yide Zhang**, Takashi Hato, Pierre C. Dagher, Evan L. Nichols, Cody J. Smith, Kenneth W. Dunn, and Scott S. Howard, "Automatic segmentation of intravital fluorescence microscopy images by K-means clustering of FLIM phasors", *Optics Letters*, vol. 44, no. 16, pp. 3928-3931, Aug. 2019, DOI: 10.1364/OL.44.003928. *Optics Letters*' Featured Paper of Volume 44, Issue 16.

J10 **Yide Zhang**†, Evan L. Nichols†, Abigail M. Zellmer, Ian H. Guldner, Cody Kankel, Siyuan Zhang, Scott S. Howard, and Cody J. Smith, "Generating intravital super-resolution movies with conventional microscopy reveals actin dynamics that construct pioneer axons", *Development*, vol. 146, no. 5, pp. dev171512, Mar. 2019, DOI: 10.1242/dev.171512. Supplementary software available at DOI: 10.7274/r0-5hhg-5578. Featured News 1 Featured News 2.

J9 **Yide Zhang**, David Benirschke, Ola Abdalsalam, and Scott S. Howard, "Generalized stepwise optical saturation enables super-resolution fluorescence lifetime imaging microscopy", *Biomedical Optics Express*, vol. 9, no. 9, pp. 4077-4093, Sept. 2018, DOI: 10.1364/BOE.9.004077.

J8 **Yide Zhang**, Prakash D. Nallathamby, Genevieve D. Vigil, Aamir A. Khan, Devon E. Mason, Joel D. Boerckel, Ryan K. Roeder, and Scott S. Howard, "Super-resolution fluorescence microscopy by stepwise optical saturation", *Biomedical Optics Express*, vol. 9, no. 4, pp. 1613-1629, Apr. 2018, DOI: 10.1364/BOE.9.001613. ***NDIIF Award for Best Biological Imaging Publication 2018***. Featured News.

J7 Genevieve Vigil, **Yide Zhang**, Aamir Khan, and Scott Howard, "Description of deep saturated excitation multiphoton microscopy for super-resolution imaging", *Journal of the Optical Society of America A*, vol. 34, no. 7, pp. 1217-1223, July 2017, DOI: 10.1364/JOSAA.34.001217.

J6 Aamir A. Khan, Genevieve D. Vigil, **Yide Zhang**, Susan K. Fullerton-Shirey, and Scott S. Howard, "Silica-coated ruthenium-complex nanoprobes for two-photon oxygen microscopy in biological media", *Optical Materials Express*, vol. 7, no. 3, pp. 1066-1076, Mar. 2017, DOI: 10.1364/OME.7.001066.

J5 **Yide Zhang**, Genevieve D. Vigil, Lina Cao, Aamir A. Khan, David Benirschke, Tahsin Ahmed, Patrick Fay, and Scott S. Howard, "Saturation-compensated measurements for fluorescence lifetime imaging microscopy", *Optics Letters*, vol. 42, no. 1, pp. 155-158, Jan. 2017, DOI: 10.1364/OL.42.000155.

J4 **Yide Zhang**, Aamir A. Khan, Genevieve D. Vigil, and Scott S. Howard, "Super-sensitivity multiphoton frequency-domain fluorescence lifetime imaging microscopy", *Optics Express*, vol. 24, no. 18, pp. 20862-20867, Sept. 2016, DOI: 10.1364/OE.24.020862.

J3 **Yide Zhang**, Aamir A. Khan, Genevieve D. Vigil, and Scott S. Howard, "Investigation of signal-to-noise ratio in frequency-domain multiphoton fluorescence lifetime imaging microscopy", *Journal of the Optical Society of America A*, vol. 33, no. 7, pp. B1-B11, July 2016, DOI: 10.1364/JOSAA.33.0000B1.

J2 Gang Bao, **Yide Zhang**, and Zhigang Zeng, "Memory analysis for memristors and memristive recurrent neural networks", *IEEE/CAA Journal of Automatica Sinica*, vol. 7, no. 1, pp. 96-105, Jan. 2020, DOI: 10.1109/JAS.2019.1911828.

J1 Shiping Wen, Zhigang Zeng, Tingwen Huang, and **Yide Zhang**, "Exponential adaptive lag synchronization of memristive neural networks via fuzzy method and applications in pseudorandom number generators", *IEEE Transactions on Fuzzy Systems*, vol. 22, no. 6, pp. 1704-1713, Dec. 2014, DOI: 10.1109/TFUZZ.2013.2294855.

SELECTED CONFERENCE PUBLICATIONS

C32 Yilin Luo, Rui Cao, **Yide Zhang**, Manxiu Cui, Yang Zhang, Konstantin Maslov, Jinge Yang, Massimo D'Apuzzo, Lihong V. Wang, "The fastest, clinically validated ultraviolet photoacoustic microscopy for label-free intraoperative tissue assessment enhanced by unsupervised deep learning", *SPIE Photonics West 2026*, San Francisco, California USA, Jan. 2026, DOI: 10.1117/12.3078865. ***Best Paper Award of Photons Plus Ultrasound Conference***.

C31 **Yide Zhang**, "High-speed optical imaging and sensing for scattering biological systems", *SPIE Photonics West 2026* (Invited), San Francisco, California USA, Jan. 2026, DOI: 10.1117/12.3082397.

C30 Yushen Zuo, Qi Zheng, Mingyang Wu, Xinrui Jiang, Renjie Li, Jian Wang, **Yide Zhang**, Gengchen Mai, Lihong V. Wang, James Zou, Xiaoyu Wang, Ming-Hsuan Yang, Zhengzhong Tu, "4KAgent: Agentic any image to 4K super-resolution", *Conference on Neural Information Processing Systems (NeurIPS 2025)*, San Diego, California USA, Dec. 2025, DOI: 10.52202/085713-5476.

C29 **Yide Zhang**, "Ultrafast photoacoustic and optical imaging for vascular and neural dynamics", *Optica Biophotonics Congress 2025* (Invited), Bio-Optics: Design and Application, San Francisco, Coronado USA, Apr. 2025, DOI: 10.1364/BODA.2025.DM3A.1.

C28 Xin Tong, Cindy Z. Liu, Yilin Luo, Li Lin, Jessica Dzubnar, Marta Invernizzi, **Yide Zhang**, Rui Cao, Peng Hu, Jaclene Torres, Armine Kasabyan, Lily L. Lai, Lisa D. Yee, and Lihong V. Wang, "Integrating machine learning with panoramic photoacoustic computed tomography for improved breast lesion analysis", *SPIE Photonics West 2025*, San Francisco, California USA, Mar. 2025, DOI: 10.1117/12.3047914.

C27 **Yide Zhang**, Peng Hu, Lei Li, Rui Cao, Anjul Khadria, Konstantin Maslov, Xin Tong, Yushun Zeng, Laiming Jiang, Qifa Zhou, and Lihong V. Wang, "Ultrafast single-shot 3D photoacoustic tomography *in vivo* using a single-element detector", *SPIE Photonics West 2024*, San Francisco, California USA, Mar. 2024, DOI: 10.1117/12.3007503. ***Seno Medical Best Paper Award***.

C26 **Yide Zhang**, Zhe He, Xin Tong, David C. Garrett, Rui Cao, and Lihong V. Wang, "Quantum imaging of biological organisms using hyperentangled photon pairs ", *SPIE Photonics West 2024*, San Francisco, California USA, Mar. 2024.

C25 Xin Tong, Zhe He, **Yide Zhang**, Lei Li, and Lihong V. Wang, "Super-resolution quantum microscopy at the Heisenberg limit", *SPIE Photonics West 2024*, San Francisco, California USA, Mar. 2023, DOI: 10.1117/12.3000759.

C24 Xin Tong, Zhe He, **Yide Zhang**, Samuel A. Solomon, Li Lin, Qiyuan Song, and Lihong V. Wang, "Experimental full-domain mapping of quantum correlation in Clauser-Horne-Shimony-Holt scenarios", *SPIE Photonics West 2024*, San Francisco, California USA, Mar. 2023, DOI: 10.1117/12.3001290.

C23 **Yide Zhang**, Binglin Shen, Tong Wu, Jerry Zhao, Joseph Jing, Peng Wang, Kanomi Sasaki-Capela, William Dunphy, David Garrett, Konstantin Maslov, Weiwei Wang, and Lihong V. Wang, "Ultrafast phase imaging of propagating current flows in myelinated axons and electromagnetic pulses in dielectrics", *SPIE Photonics West 2023*, San Francisco, California USA, Feb. 2023, DOI: 10.1117/12.2653-137. ***Hitachi High-Tech Best Presentation Award***.

C22 Xin Tong, Li Lin, Susana Cavallero, **Yide Zhang**, Shuai Na, Rui Cao, Tzung K. Hsiai, and Lihong V. Wang, "Non-invasive photoacoustic computed tomography of cardiac anatomy and function in rats", *SPIE Photonics West 2023*, San Francisco, California USA, Feb. 2023, DOI: 10.1117/12.2653147.

C21 Rui Cao, Jingjing Zhao, Lei Li, Lin Du, **Yide Zhang**, Yilin Luo, Laiming Jiang, Samuel P. Davis, Qifa Zhou, Adam de la Zerda, and Lihong V. Wang, "Needle-shaped beam optical-resolution photoacoustic microscopy with an extended depth of field", *SPIE Photonics West 2023*, San Francisco, California USA, Feb. 2023, DOI: 10.1117/12.2650561.

C20 Rui Cao, Scott D. Nelson, Samuel Davis, Yu Liang, Yilin Luo, **Yide Zhang**, Brooke Crawford, and Lihong V. Wang, "Label-free ultraviolet photoacoustic histology via deep learning for rapid intraoperative diagnosis of bone cancer", *SPIE Photonics West 2022*, San Francisco, California USA, Feb. 2022. ***Best Paper Award of Photons Plus Ultrasound Conference***.

C19 Varun Mannam, **Yide Zhang**, Xiaotong Yuan, Takashi Hato, Pierre C. Dagher, Evan L. Nichols, Cody J. Smith, Kenneth W. Dunn, and Scott Howard, "Convolutional neural network denoising in fluorescence lifetime imaging microscopy (FLIM)", *SPIE Photonics West 2021*, Online Only, Mar. 2021, DOI: 10.1117/12.2578574. ***BiOS'21 3-Minute Poster Prize***.

C18 Varun Mannam, **Yide Zhang**, Xiaotong Yuan, and Scott Howard, "Deep learning-based super-resolution fluorescence microscopy on small datasets", *SPIE Photonics West 2021*, Online Only, Mar. 2021, DOI: 10.1117/12.2578519.

C17 Xiaotong Yuan, Varun Mannam, **Yide Zhang**, and Scott Howard, "Overcoming the fundamental limitation of frequency-domain fluorescence lifetime imaging microscopy spatial resolution", *SPIE Photonics West 2021*, Online Only, Mar. 2021, DOI: 10.1117/12.2577284.

C16 Varun Mannam, **Yide Zhang**, Yinhao Zhu, and Scott Howard, "Instant image denoising plugin for ImageJ using convolutional neural networks", *Biomedical Optics 2020*, Washington, DC USA, Apr. 2020, DOI: 10.1364/MICROSCOPY.2020.MW2A.3.

C15 **Yide Zhang**†, Yinhao Zhu†, Evan Nichols, Qingfei Wang, Siyuan Zhang, Cody Smith, and Scott Howard, "A Poisson-Gaussian denoising dataset with real fluorescence microscopy images", *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2019)*, Long Beach, California USA, June 2019, DOI: 10.1109/CVPR.2019.01198.

C14 **Yide Zhang**, Ian H. Guldner, Evan L. Nichols, David Benirschke, Cody J. Smith, Siyuan Zhang, and Scott S. Howard, "Three-dimensional deep tissue multiphoton frequency-domain fluorescence lifetime imaging microscopy via phase multiplexing and adaptive optics", *SPIE Photonics West 2019*, San Francisco, California USA, Feb. 2019, DOI: 10.1117/12.2510674. ***JenLab Young Investigator Award***. Featured News.

C13 **Yide Zhang**, David Benirschke, Ola Abdalsalam, and Scott S. Howard, "Super-resolution multiphoton frequency-domain fluorescence lifetime imaging microscopy by generalized stepwise optical saturation (GSOS)", *SPIE Photonics West 2019*, San Francisco, California USA, Feb. 2019, DOI: 10.1117/12.2507663.

C12 Ola Abdalsalam, **Yide Zhang**, Scott Howard, and Thomas D. O'Sullivan, "Self-calibrated frequency domain diffuse optical spectroscopy with a phased source array", *SPIE Photonics West 2019*, San Francisco, California USA, Feb. 2019, DOI: 10.1117/12.2510422.

C11 **Yide Zhang**, Prakash D. Nallathamby, Evan L. Nichols, Cody J. Smith, Ryan K. Roeder, and Scott S. Howard, "Super-resolution and high-speed quantitative multiphoton microscopy", *Seventh Annual Harper Cancer Research Institute Research Day*, Notre Dame, Indiana USA, Apr. 2018, DOI: 10.7274/r0-zqes-kr15. ***Poster Contest Third Prize***.

C10 **Yide Zhang**, David Benirschke, and Scott S. Howard, "Stepwise optical saturation microscopy: obtaining super-resolution images with conventional fluorescence microscopes", *Biomedical Optics 2018*, Hollywood, Florida USA, Apr. 2018, DOI: 10.1364/TRANSLATIONAL.2018.JTh3A.27.

C9 **Yide Zhang**, David Benirschke, and Scott S. Howard, "Super-resolution fluorescence imaging by stepwise optical saturation microscopy", *IEEE Annual Mini-symposium on Electron Devices and Photonics*, Notre Dame, Indiana USA, Oct. 2017, DOI: 10.7274/r0-bbyt-9j45. ***Best Presentation Award***.

C8 **Yide Zhang**, Genevieve D. Vigil, Aamir A. Khan, and Scott S. Howard, "Doubling the sensitivity of multiphoton frequency-domain fluorescence lifetime images", *CLEO: Science and Innovations 2017*, San Jose, California USA, May 2017, DOI: 10.1364/CLEO_SI.2017.SM3C.6.

C7 Genevieve D. Vigil, **Yide Zhang**, Aamir A. Khan, and Scott S. Howard, "Simulation and experimental design of saturated excitation (SAX) multiphoton microscopy (MPM)", *CLEO: Applications and Technology 2017*, San Jose, California USA, May 2017, DOI: 10.1364/CLEO_AT.2017.JTu5A.64.

C6 **Yide Zhang**, Genevieve D. Vigil, Aamir A. Khan, and Scott S. Howard, "Super-sensitivity and super-resolution multiphoton fluorescence lifetime imaging microscopy", *NDnano Symposium: Nanotechnology in the Treatment of Neurodegenerative Disorders*, Notre Dame, Indiana USA, Mar. 2017, DOI: 10.7274/r0-k1dc-5q52. ***Best Poster Award***.

C5 Aamir A. Khan, Susan K. Fullerton-Shirey, Genevieve D. Vigil, **Yide Zhang**, and Scott S. Howard, "Highly stable two-photon oxygen imaging probe based on a ruthenium-complex encapsulated in a silica-coated nanomicelle", *CLEO: Applications and Technology 2016*, San Jose, California USA, June 2016, DOI: 10.1364/CLEO_AT.2016.ATu4O.3.

C4 Aamir A. Khan, Genevieve D. Vigil, **Yide Zhang**, and Scott S. Howard, "Theoretical analysis of the signal-to-noise ratio of two-photon oxygen imaging probes", *Biomedical Optics 2016*, Fort Lauderdale, Florida USA, Apr. 2016, DOI: 10.1364/CANCER.2016.JW3A.48.

C3 **Yide Zhang**, Genevieve D. Vigil, Aamir A. Khan, and Scott S. Howard, "High-resolution frequency-domain multiphoton fluorescence lifetime imaging microscopy", *Fifth Annual Harper Cancer Research Institute Research Day*, Notre Dame, Indiana USA, Apr. 2016, DOI: 10.7274/r0-vt1g-bp12.

C2 **Yide Zhang** and Scott S. Howard, "On increasing the imaging rate of frequency-domain multiphoton fluorescence lifetime imaging microscopy", *IEEE Annual Mini-symposium on Electronic and Photonics*, Notre Dame, Indiana USA, Oct. 2015, DOI: 10.7274/r0-dw42-x854.

C1 **Yide Zhang**, Zhigang Zeng, and Shiping Wen, "Implementation of memristive neural networks with spike-rate-dependent plasticity synapse", *IEEE International Joint Conference on Neural Networks (IJCNN)*, Beijing, China, July 2014, DOI: 10.1109/IJCNN.2014.6889740.

PATENTS

P7 Lihong Wang, Xin Tong, Chien-Ying Huang, **Yide Zhang**, "Quantum imaging with parallel sources and detection", US Provisional Patent Application (filed on May 16, 2025).

P6 Lihong Wang, Xin Tong, Zhe He, **Yide Zhang**, "Hyper-Heisenberg scaling quantum microscopy", US Patent Application Number: US20250093636A1 (filed on Sept. 20, 2024).

P5 Lihong Wang, **Yide Zhang**, "Single-shot 3D imaging using a single detector", US Patent Application Number: US20240241239A1 (filed on Jan. 11, 2024).

P4 Scott Howard, Genevieve Vigil, and **Yide Zhang**, "Super-sensitivity multiphoton frequency-domain fluorescence lifetime imaging microscopy", US Patent Number: US11181727B2 (granted on Nov. 23, 2021).

P3 Scott Howard, **Yide Zhang**, and Cody J. Smith, "Super-resolution fluorescence microscopy by stepwise optical saturation", US Patent Number: US11131631B2 (granted on Sept. 28, 2021).

P2 **Yide Zhang**, Zhigang Zeng, Yidong Zhu, Mingfu Cao, and Junfeng Zhao, "Signal processing circuit", US Patent Number: US10586590B2 (granted on Mar. 10, 2020), EP Patent Number: EP3282449B1 (granted on Aug. 7, 2019), CN Patent Number: CN107210064B (granted on Feb. 14, 2020).

P1 **Yide Zhang**, Zhigang Zeng, Shiping Wen, Mingfu Cao, and Junfeng Zhao, "Neuron simulation circuit", CN Patent Application Number: CN201510508806.6A, Publication Number: CN106470023A.

GRANTS

NIH K99/R00 Pathway to Independence Award 2024-2028

- Sponsor: National Institute of Biomedical Imaging and Bioengineering (NIBIB), National Institutes of Health (NIH)
- Title: "Next-generation photoacoustic imaging for real-time, non-invasive monitoring of brain oxygen metabolism"
- Role: Principal Investigator
- Budget: $874,465
- Duration: 48 months

NIH T32 Ruth L. Kirschstein Institutional National Research Service Award 2023-2024

- Sponsor: National Heart, Lung, and Blood Institute (NHLBI), National Institutes of Health (NIH)
- Title: "Single-shot 3D photoacoustic tomography using a single-element detector for cardiovascular imaging"
- Role: Principal Investigator
- Total budget: $78,613
- Duration: 12 months

S2I Early- or Mid-Stage Research Projects 2023-2025

- Sponsor: Center for Sensing to Intelligence (S2I), Caltech
- Title: "N-fold super-resolution quantum microscopy"
- Role: Co-Investigator
- Total budget: $90,000
- Duration: 24 months

Berry Family Foundation Graduate Fellowship 2017-2018

- Sponsor: Advanced Diagnostics and Therapeutics (AD&T), University of Notre Dame
- Title: "Fast, accurate, and noninvasive diagnostic and therapeutic techniques enabled by super-sensitivity, super-resolution, and super-penetration quantitative multiphoton microscope in living tissue"
- Role: Principal Investigator
- Total budget: $25,000
- Duration: 12 months

SELECTED AWARDS AND HONORS

2024	**NIH K99/R00 Pathway to Independence Award**, National Institute of Biomedical Imaging and Bioengineering (NIBIB), NIH
2024	**Seno Medical Best Paper Award**, Photons Plus Ultrasound: Imaging and Sensing Conference, SPIE Photonics West
2023	**Hitachi High-Tech Best Presentation Award**, High-Speed Biomedical Imaging and Spectroscopy Conference, SPIE Photonics West
2022	**IOP Publishing Outstanding Reviewer Award**, IOP Publishing, Bristol UK
2021	**IOP Publishing Outstanding Reviewer Award**, IOP Publishing, Bristol UK
2021	**OSA Outstanding Reviewer**, The Optical Society, Washington DC USA
2021	**BiOS'21 3-Minute Poster Prize**, SPIE Photonics West
2019	**IOP Publishing Outstanding Reviewer Award**, IOP Publishing, Bristol UK
2019	**NDIIF Best Biological Imaging Publication Award 2018**, Featured News Notre Dame Integrated Imaging Facility (NDIIF), Indiana USA
2019	**CRC Award for Computational Sciences and Visualization**, Featured News Center for Research Computing (CRC), University of Notre Dame, Indiana USA
2019	**JenLab Young Investigator Award**, Featured News Multiphoton Microscopy in the Biomedical Sciences Conference, SPIE Photonics West
2018	**James L. Massey Travel Grant**, Department of Electrical Engineering, University of Notre Dame, Indiana USA
2018	**Poster Contest Third Prize**, Seventh Annual Harper Cancer Research Institute Research Day, Indiana USA
2017	**Best Presentation Award**, IEEE Annual Mini-symposium on Electron Devices and Photonics, Indiana USA
2017	**Berry Family Foundation Graduate Fellowship**, Featured News Advanced Diagnostics and Therapeutics (AD&T), Indiana USA

2017	**Graduate Student Research Award** (One Awardee at NDEE), Department of Electrical Engineering, University of Notre Dame, Indiana USA
2017	**Best Poster Award**, NDnano Symposium, University of Notre Dame, Indiana USA
2014	**Outstanding Undergraduate Thesis Award** (Ranking 1/205), Huazhong University of Science and Technology (HUST), Wuhan, China
2013	**Meritorious Winner**, Mathematical Contest in Modeling (MCM), Consortium for Mathematics and Its Applications (COMAP)
2013	**Decent Capital's Scholarship** (Top 0.1%), Decent Capital
2013	**National Scholarship** (Top 1%), Ministry of Education of China
2012	**National Scholarship** (Top 1%), Ministry of Education of China
2011	**National Scholarship** (Top 1%), Ministry of Education of China
2012	**Prominent Student Award** (Top 1%), HUST, Wuhan, China

MEDIA COVERAGE

- "New Quantum Microscopy Trick Quadruples Microscope Resolution", Caltech News, Aug. 6, 2026.
- "Four ECEE Students Earn Major National Science Foundation Research Fellowships", CU Boulder ECEE News, May 7, 2026.
- "Prestigious NSF Fellowship Awarded to 40 Graduate Students", CU Boulder Graduate School News, Apr. 14, 2026.
- "Intraoperative Tumor Histology May Enable More-Effective Cancer Surgeries", Caltech News, Jan. 12, 2026.
- "Three New Faculty Members Joining the BME Department for Fall 2025", CU Boulder BME News, Sept. 8, 2025.
- "Meet Assistant Professor Yide Zhang", CU Boulder ECEE News, Aug. 11, 2025.
- "ECEE Welcomes New Faculty for Fall 2025", CU Boulder ECEE News, Aug. 11, 2025.
- "AI-Assisted Technique Offers Safe, Effective, Painless Breast Imaging Alternative", Caltech News, June 30, 2025.
- "New Microscopy Technique Could Enable Rapid Tumor Analysis in the Operating Room", Caltech News, Dec. 18, 2024.
- "Using Polarization to Improve Quantum Imaging", Caltech News, Mar. 19, 2024.
- "New Technology Brings Advanced Blood Imaging Closer to the Clinic", NIH NIBIB's *Science Highlights*, Feb. 16, 2024.
- "Fast Capturing of Deep Blood Flow", *Nature Biomedical Engineering*'s News & Views, Dec. 27, 2023.
- "*2023 Year in Review*", Caltech News, Dec. 18, 2023.
- "Advancements Make Laser-Based Imaging Simpler and Three-Dimensional", Caltech News, Dec. 1, 2023.
- "A Quantum Entanglement Microscope", The Science Blog of Fédération Française de Sociétés Scientifiques, July 3, 2023.
- "Quantum Entanglement Doubles Microscope Resolution", Physics World, June 6, 2023.
- "Quantum Entanglement of Photons Doubles Microscope Resolution", Caltech News, May 1, 2023.
- "Worlds Fastest Laser Camera Films Combustion in Real Time", University of Gothenburg News, Feb. 24, 2023.
- "Wavefront Shaping: From Telescopes to Biological Tissue", Caltech News, Feb. 24, 2023.
- "Sharp Laser Beam Reveals Internal Organs in Stunning 3D", *Nature* Research Highlights, Dec. 6, 2022.
- "Seeing More with a Needle-Shaped Laser", Caltech News, Dec. 1, 2022.
- "Super-Fast Camera Captures Electrical Signals Moving Through Nerve Cells", PetaPixel, Oct. 11, 2022.
- "High-Speed Camera Captures Signals Traveling Through Nerve Cells", Caltech News, Oct. 6, 2022.

- "Laser Light Offers New Tool for Treating Bone Cancer", Caltech News, Sept. 19, 2022.
- "Notre Dame Professor Discovers New Cell Imaging Technique", The Observer, Oct. 26, 2020.
- "Super Resolution Ghosts", University of Notre Dame Stories, Oct. 1, 2019.
- "Bringing Super-Resolution Microscopy to the Masses", Wiley Analytical Science, Sept. 11, 2019.
- "Your Motivation Matters: Engineering to Heal", The Wire at the University of Notre Dame, Sept. 10, 2019.
- "NDIIF Celebrates 10 Year Anniversary at Annual Imaging Workshop", Biophysics at Notre Dame, May 17, 2019.
- "Super Resolution Imaging Made Easy", Wiley Analytical Science, Mar. 5, 2019.
- "Open-Source Application Creates Super-Resolution Images of Cell Development in Living Animals", Notre Dame News, Mar. 1, 2019.
- "Graduate Student Receives JenLab Young Investigator Award", Notre Dame Research, Feb. 18, 2019.
- "A "Simple" Solution to a Common Situation", The College of Engineering at the University of Notre Dame, Apr. 5, 2018.
- "Notre Dame Graduate Students Awarded Fellowships for Cross-Disciplinary Biomedical Research", Notre Dame Research, July 5, 2017.

TEACHING AND MENTORING

Courses

- ECEN 4006/5006: Biomedical Optics, University of Colorado Boulder, Fall 2026
- BMEN 3030: Bioinstrumentation, University of Colorado Boulder, Spring 2026
- ECEN 5006: Biomedical Optical Imaging, University of Colorado Boulder, Fall 2025
- EE 30342: Microelectronic Circuit Design, University of Notre Dame, Spring 2015

Current Students

- Jacob Stewart, ECEE PhD Student, July 2025–Present
  - **NSF Graduate Research Fellowship Program (GRFP) Fellow**, 2026
- Qingye Hu, ECEE PhD Student, Aug. 2025–Present
- Hengyue Zhu, ECEE PhD Student, Aug. 2025–Present
- Wanqing Zhang, BME PhD Student, Jan. 2026–Present
- Jeerakit Kanokthipphayakun, ECEE Master Student, Aug. 2025–Present
- Jonathan Knee, ECEE Master Student, Nov. 2025–Present
- Trejan Schuneman, ECEE Undergraduate Student, Apr. 2026–Present
- Kamila Skonieczny, BME Undergraduate Student, May 2026–Present
- Brennan Grady, BME Undergraduate Student, May 2026–Present

Alumni

- Souparna Basu, Physics Research Assistant, May 2025–Apr. 2026 (now at the University of Tennessee, Knoxville)

Thesis Committees

- William Frantz, Mark Borden Lab, University of Colorado Boulder, USA, 2026
- Darwin Quiroz, Juliet Gopinath Lab, University of Colorado Boulder, USA, 2026
- Yiming Lai, Jinyang Liang Lab, Institut national de la recherche scientifique (INRS), Canada, 2025

Programs

- Transforming Your Research into Teaching, California Institute of Technology, 2023
- Teaching Statement Workshop, California Institute of Technology, 2022

- Teaching Well Using Technology, University of Notre Dame, 2018
- Clay International Academy Outreach Program, University of Notre Dame, 2016

PROFESSIONAL SERVICE

Invited Speaker

- “Exploring Biomedical Optical Imaging”, *CU Boulder BMEN 1000 Exploring Biomedical Engineering Seminar*, University of Colorado Boulder, Boulder, Colorado USA, Mar. 30, 2026.
- “High-Speed Optical Imaging and Sensing for Scattering Biological Systems”, *SPIE Photonics West*, San Francisco, California USA, Jan. 17, 2026.
- “Advancing Biomedical Optical Imaging for Discovery and Health”, *CU Boulder BME Seminar Series*, University of Colorado Boulder, Boulder, Colorado USA, Sept. 5, 2025.
- “Ultrafast Photoacoustic and Optical Imaging for Vascular and Neural Dynamics”, *Optica Biophotonics Congress: Optics in the Life Sciences*, Coronado, California USA, Apr. 21, 2025.
- “Transitioning to Independent Research in Optical Imaging”, *NIH Caltech/UCLA T32 Symposium*, Los Angeles, California USA, Apr. 3, 2025.
- “Single-Shot 3D Photoacoustic Imaging using a Single-Element Detector”, *22nd Annual NeuroTech Convention of the Society for Brain Mapping and Therapeutics (SBMT)*, Los Angeles, California USA, Feb. 28, 2025.
- “Extreme Imaging: Breaking the Fundamental Barriers in Optical Imaging”, *Wuhan National Laboratory for Optoelectronics Biomedical Young Scholar Seminar*, Huazhong University of Science and Technology, Wuhan, Hubei China, Nov. 28, 2024.
- “Extreme Imaging: Breaking the Fundamental Barriers in Optical Imaging”, *School of Artificial Intelligence and Automation AIA Frontier Seminar*, Huazhong University of Science and Technology, Wuhan, Hubei China, Nov. 25, 2024.
- “From T32 to K99: My Path to Independence in Optical Imaging”, *NIH Caltech/UCLA T32 Symposium*, Pasadena, California USA, Sept. 19, 2024.
- “Q&A Panel on Faculty Applications”, *Caltech Postdoctoral Association*, Pasadena, California USA, Aug. 23, 2024.
- “Extreme Imaging: Breaking the Fundamental Barriers in Optical Imaging”, *Special Seminar of Wyant College of Optical Sciences*, The University of Arizona, Tucson, Arizona USA, Apr. 16, 2024.
- “Advancing the Frontiers of AMO Physics: Extreme Imaging Techniques for Optical and Quantum Sciences”, *TAMU AMO/IQSE Seminar*, Texas A&M University, College Station, Texas USA, Apr. 5, 2024.
- “Extreme Imaging: Surpassing Optical Limits for Scientific Discovery”, *TAMU Physics and Astronomy Colloquium*, Texas A&M University, College Station, Texas USA, Apr. 4, 2024.
- “Extreme Imaging: Breaking the Fundamental Barriers in Optical Imaging”, *UC Irvine EECS and BLI Seminar*, University California, Irvine, Irvine, California USA, Mar. 13, 2024.
- “Ultrafast Single-Shot 3D Photoacoustic Tomography *In Vivo* using a Single-Element Detector”, *A Special MedE Symposium to Celebrate the 10th Anniversary of the Andrew and Peggy Cherng Department of Medical Engineering*, Pasadena, California USA, Mar. 7, 2024.
- “Extreme Imaging: Breaking the Fundamental Barriers in Optical Imaging”, *CU Boulder ECEE Seminar*, University of Colorado Boulder, Boulder, Colorado USA, Feb. 19, 2024.
- “Extreme Imaging: Breaking the Fundamental Barriers in Optical Imaging”, *UT Austin ECE Seminar*, The University of Texas at Austin, Austin, Texas USA, Feb. 12, 2024.
- “Ultrafast Single-Shot 3D Photoacoustic Tomography *In Vivo* using a Single-Element Detector”, *UCLA Cardiovascular Engineering Research Laboratory (PI: Tzung Hsiai)*, UCLA, Online, Jan. 26, 2024.
- “Ultrafast Single-Shot 3D Photoacoustic Tomography *In Vivo* using a Single-Element Detector”, *The 15th International Conference on Ultrasound Engineering for Biomedical Applications*, Torrance, California USA, July 21, 2023.

- "Ultrafast Phase Imaging of Propagating Current Flows in Myelinated Axons and Electromagnetic Pulses in Dielectrics", *Caltech Postdoc L(a)unch Seminar*, Pasadena, California USA, May 5, 2023.
- "Ultrafast and Hypersensitive Phase Imaging of Propagating Internodal Current Flows in Myelinated Axons", *The International Society for Brain Mapping and Therapeutics (SBMT) 2023/20th Annual World Congress of SBMT*, Los Angeles, California USA, Feb. 16, 2023.
- "Ultrafast Phase Imaging of Propagating Current Flows in Myelinated Axons and Electromagnetic Pulses in Dielectrics", *Complex Media Optics Lab (PI: Sylvain Gigan), École Normale Supérieure (ENS)*, Online, Dec. 6, 2022.
- "Ultrafast Phase Imaging of Propagating Current Flows in Myelinated Axons and Electromagnetic Pulses in Dielectrics", *Caltech Electrical Engineering Advisory Council Meeting*, Pasadena, California USA, Nov. 29, 2022.
- "Ultrafast Phase Imaging of Propagating Current Flows in Myelinated Axons and Electromagnetic Pulses in Dielectrics", *Dushu Forum/The 2nd Research Forum for Young Scientists, University of Science and Technology of China*, Online, Nov. 25, 2022.
- "NDCSSA Excellent Graduate Experience Sharing Session", *Notre Dame Chinese Students and Scholars Association (NDCSSA) Talk No. 3*, Notre Dame, Indiana USA, Aug. 28, 2019.
- "NDIIF Best Biological Imaging Publication Award 2018: Stepwise Optical Saturation (SOS) Microscopy and Beyond", *6th Annual Midwest Imaging and Microanalysis Workshop*, Notre Dame, Indiana USA, May 6, 2019.
- "Three-Dimensional, Deep Tissue, Super-Resolution Multiphoton FLIM via Phase Multiplexing", *AD&T External Review*, Notre Dame, Indiana USA, Mar. 25, 2019.
- "Research in ND Electrical Engineering: A Graduate Student's Perspective", *Notre Dame Electrical Engineering Graduate Applicant Recruiting Event*, Notre Dame, Indiana USA, Mar. 8, 2019.
- "Novel Super-Resolution Fluorescence Microscopy Methods in Application to Bio and Materials Sciences", *Notre Dame Electron Microscopy Club Meeting*, Notre Dame, Indiana USA, Feb. 13, 2019.

Editorship

- Editorial Board Member, *Journal of Optics and Photonics Research*
- Guest Editor, Special Issue "Quantum Imaging", *Advanced Imaging*
- Guest Editor, Special Issue "Advanced Optical Detection and Imaging Systems", *Electronics*
- Topic Editor, Research Topics "Phasor Analysis for Fluorescence Lifetime Data", *Frontiers in Bioinformatics*
- Guest Associate Editor, Cellular Biochemistry Section, *Frontiers in Cell and Developmental Biology*
- Guest Editor, Special Issue "Interpretation of Machine Learning: Prediction, Representation, Modeling, and Visualization", *Computational Intelligence and Neuroscience*
- Guest Editor, Special Issue "Advanced Optics Engineering", *Materials*

Journal Reviewer

- **Nature Portfolio**: Nature Methods, Nature Biomedical Engineering, Nature Computational Science, Nature Communications, Light: Science & Applications, npj Imaging, Communications Engineering, Communications Physics, Scientific Reports
- **AAAS**: Science Advances
- **Cell Press**: The Innovation, Cell Reports Methods
- **Optica (OSA)**: Optica, Optica Quantum, Photonics Research, Biomedical Optics Express, Optics Letters, Optics Express, Journal of the Optical Society of America A, Applied Optics, Optics Continuum
- **SPIE**: Journal of Biomedical Optics, Advanced Photonics, Advanced Photonics Nexus, Neurophotonics, Journal of Electronic Imaging, Optical Engineering, Journal of Applied Remote Sensing
- **IEEE**: IEEE Electron Device Letters, IEEE Transactions on Biomedical Engineering, IEEE Photonics Technology Letters, IEEE Transactions on Smart Grid

- **IOP**: Physics in Medicine and Biology, Methods and Applications in Fluorescence, Journal of Physics: Photonics, Journal of Optics, Inverse Problems, Engineering Research Express, New Journal of Physics, Machine Learning: Science and Technology, Physica Scripta, Measurement Science and Technology, Biomedical Physics & Engineering Express, Quantum Science and Technology
- **AIP**: Applied Physics Letters, Journal of Applied Physics, Review of Scientific Instruments, APL Photonics, APL Machine Learning, AIP Advances, Physics of Fluids, AVS Quantum Science, Biophysics Reviews
- **Elsevier**: Photoacoustics, Knowledge-Based Systems, Pattern Recognition Letters, Optics Communications, Infrared Physics and Technology, Sensors and Actuators A: Physical, Optics and Laser Technology, Journal of Hazardous Materials Advances
- **MDPI**: Bioengineering, Sensors, Photonics, Diagnostics, Applied Sciences, Life, Micromachines, Electronics, Mathematics, Information, Cancers, Veterinary Sciences, Biomolecules, Machine Learning and Knowledge Extraction, Computers, Pharmaceutics, Journal of Imaging, Current Oncology, Radiation, Biomimetics
- **Frontiers**: Frontiers in Physiology, Frontiers in Computer Science, Frontiers in Bioinformatics, Frontiers in Molecular Biosciences, Frontiers in Earth Science
- **Springer**: eLight, Food and Bioprocess Technology, Cytotechnology, BioMedical Engineering OnLine
- **Wiley**: Journal of Biophotonics
- **ACS**: ACS Photonics

Conference Reviewer

- IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)
- Conference on Neural Information Processing Systems (NeurIPS)
- European Conference on Computer Vision (ECCV)
- International Symposium on Computational Sensing (ISCS)
- International Conference on Intelligent Control and Information Processing (ICICIP)

Grant Reviewer

- DOE Office of Science Graduate Student Research (SCGSR)
- NSF Cyberinfrastructure for Sustained Scientific Innovation (CSSI)
- ARL Broad Agency Announcement (BAA)
- Research Grants Council (RGC) of Hong Kong General Research Fund (GRF)
- Berthiaume Institute For Precision Health Technology Development Fund
- Berthiaume Institute For Precision Health Discovery Fund
- Institute for Precision Health at Notre Dame Fund

Committee Member

- BME Undergraduate Committee, 2025–Present

Conference Session Chair

- Session Chair of Novel Biomedical Optical Technology II, *Optica Biophotonics Congress: Optics in the Life Sciences 2025*.

Contributor to Research Community

- Medical Diagnostics, *Nature Outlook*, Dec. 13, 2023.
- "Single-shot 3D photoacoustic tomography based on a single-element detector for ultrafast imaging of hemodynamics", Bioengineering and Biotechnology at Nature Research, Dec. 4, 2023.
- IOP Trusted Reviewer, IOP Publishing, Sept. 22, 2020.
- Certified Publons Academy Peer Reviewer, Publons, June 8, 2020.

- Certified Reviewer for OSA Journals, The Optical Society (OSA), Dec. 15, 2019.
- " 'Publish or perish' will not perish", Behavioural and Social Sciences at Nature Research, Oct. 10, 2019.

PROFESSIONAL MEMBERSHIP

Member, Optica–formerly the Optical Society (OSA)
Member, International Society for Optics and Photonics (SPIE)
Member, Institute of Electrical and Electronics Engineers (IEEE)
Member, American Association for the Advancement of Science (AAAS)